

Credentials, API Keys & IDs Required

North Rift GeoAI + Agentic EWS — Setup Reference for the Production Codebase
Maps directly to environment variables in `.env.example`

How to Use This Document

Every external service the codebase calls is listed below with what credential is needed, where to obtain it, which environment variable it maps to, and which module depends on it. Nothing in the code is hard-coded — copy `.env.example` to `.env` and fill in the values as you obtain them. Items are grouped by priority:

- **BLOCKING** — the corresponding module will raise an error immediately without this credential.
- **IMPORTANT** — needed for full functionality/production reliability, but the module degrades gracefully or has a fallback without it.
- **OPTIONAL** — public endpoints that work without a key, or alternate/enhancement paths.

1. Earth Observation & GeoAI Sources

Priority	Service	How to obtain	Env variable(s)	Used in
BLOCKING	Copernicus Data Space Ecosystem (Sentinel-1/2, GLO-30 DEM)	Register at dataspace.copernicus.eu ; create an OAuth2 client to obtain a Client ID and Client Secret	COPERNICUS_CLIENT_ID COPERNICUS_CLIENT_SECRET	geoai/dem_terrain.py
IMPORTANT	Earth Genome — Earth Index	Sign up at app.earthindex.ai for an account. No confirmed public API key was documented at time of writing — contact Earth Genome (info@earthgenome.org) to confirm current API/auth model	EARTH_INDEX_APP_ACCOUNT_EMAIL EARTH_INDEX_API_KEY	geoai/earthindex_client.py
OPTIONAL	Earth Index bulk embeddings (Source Cooperative)	Public GeoParquet archive at source.coop/earthgenome/earthindexembeddings ; token only needed if bucket policy requires it in future	EARTH_INDEX_SOURCE_COOP_TOKEN	geoai/earthindex_client.py
IMPORTANT	NASA Earthdata (MODIS via LP DAAC, GPM IMERG)	Create a free account at urs.earthdata.nasa.gov	NASA_EARTHDATA_USERNAME NASA_EARTHDATA_PASSWORD	ews/data_fetch_agent.py
OPTIONAL	Google Earth Engine (alternate MODIS/Sentinel path)	Create a GEE service account in Google Cloud Console; download the JSON key	GEE_SERVICE_ACCOUNT_EMAIL GEE_SERVICE_ACCOUNT_KEY_PATH	ews/data_fetch_agent.py
OPTIONAL	NASA POWER (meteorological variables)	Public API — no credentials required	— none —	ews/data_fetch_agent.py
OPTIONAL	CHIRPS / TAMSAT (rainfall)	Public data services (IRI Data Library / Climate Hazards Center) — no credentials required for standard endpoints	— none —	ews/data_fetch_agent.py
IMPORTANT	ICPAC (seasonal outlook)	Public bulletins available without a key; a partner API key may unlock structured data — contact ICPAC directly to request one	ICPAC_API_KEY	ews/data_fetch_agent.py

Priority	Service	How to obtain	Env variable(s)	Used in
IMPORTANT	TAHMO (in-situ weather stations)	Register at tahmo.org and request API credentials	TAHMO_API_USERNAME TAHMO_API_PASSWORD	ews/data_fetch_agent.py

2. Water Resources & Conflict/Crime Data

Priority	Service	How to obtain	Env variable(s)	Used in
BLOCKING	Kenya WRA WRIS (Water Resources Information System)	Request API access directly from the Water Resources Authority (WRA); typically requires an institutional data-sharing request	WRA_WRIS_API_KEY WRA_WRIS_CLIENT_ID	geoai/borehole_density.py
IMPORTANT	Water Point Data Exchange (WPDx)	Public Socrata endpoint works unauthenticated at low volume; register at waterpointdata.org for an app token to raise rate limits	WPDX_API_KEY	geoai/borehole_density.py
OPTIONAL	ACLED (conflict event data)	Register at acleddata.com/register with an email; API key issued after approval	ACLED_API_KEY ACLED_EMAIL	Referenced by crime-dashboard companion project
OPTIONAL	KNBS / Kenya National Police Service reports	No API — obtained as periodic published reports/bulletins; ingest manually or via scheduled document parsing	— none —	Referenced by crime-dashboard companion project

3. Farmer Messaging (WhatsApp / SMS)

Priority	Service	How to obtain	Env variable(s)	Used in
BLOCKING	Meta WhatsApp Business Cloud API	Create a Meta Developer account -> create an App -> add the WhatsApp product -> add/verify a phone number -> generate a permanent access token	META_WHATSAPP_PHONE_NUMBER_ID META_WHATSAPP_BUSINESS_ACCOUNT_ID META_WHATSAPP_ACCESS_TOKEN META_APP_ID META_APP_SECRET	messaging/whatsapp_client.py
IMPORTANT	Twilio (SMS fallback / alternate WhatsApp sender)	Create a Twilio account at twilio.com ; provision a phone number or WhatsApp sender	TWILIO_ACCOUNT_SID TWILIO_AUTH_TOKEN TWILIO_FROM_NUMBER	messaging/sms_client.py

4. Infrastructure

Priority	Service	How to obtain	Env variable(s)	Used in
BLOCKING	PostGIS-enabled PostgreSQL / TimescaleDB	Provision a Postgres instance with the PostGIS extension enabled (<code>CREATE EXTENSION IF NOT EXISTS postgis;</code>); managed options include Supabase, Render, AWS RDS, or a self-hosted instance	DATABASE_URL	db/database.py, all agents persisting state

5. Areas of Interest (AOI) Requiring Configuration (Not Credentials)

Beyond credentials, the following AOI-specific values must be supplied before the pipeline runs correctly for the North Rift — these are not secrets, but they are required inputs:

AOI item	Location in code	What's needed
County boundary GeoJSON files	geoai/boundaries.py	Place verified boundary files in data/boundaries/: nandi.geojson, uasin_gishu.geojson, trans_nzoia.geojson, elgeyo_marakwet.geojson, west_pokot.geojson
Copernicus DEM bounding box	COPERNICUS_DEM_BBOX env var	Pre-set to the Nandi bbox (34.74,-0.108,35.44,0.563) — extend to cover the full North Rift AOI before regional rollout
Earth Index / UTM tile codes	geoai/earth_index_client.py: utm_tiles_for_north_rift()	Placeholder tile codes — verify against the actual Earth Index / Source Cooperative UTM 36N tiling scheme for the five counties
Earth Index reference signatures	geoai/anomaly_scoring.py	Analyst must manually flag reference tiles for each stress pattern (Kingwal Wetland degradation, stressed maize/wheat fields, bare-soil borehole pads) before anomaly scoring is meaningful
Plot-to-cell key assignment logic	ews/onboarding_agent.py: _assign_earth_index_tile / _assign_dem_cell / _assign_borehole_cell	Placeholder lon/lat-based keys — replace with the confirmed tiling/grid scheme once finalized

6. Quick-Start Checklist

- Copy `.env.example` to `.env`.
- Obtain the four BLOCKING credentials first: Copernicus OAuth2 client, WRA WRIS API key, Meta WhatsApp Business credentials, and a PostGIS DATABASE_URL — the pipeline and orchestrator will raise clear errors naming any that are still missing.
- Provide verified county boundary GeoJSON files for all five North Rift counties.
- Confirm the Earth Index API/auth model directly with Earth Genome, or proceed with the bulk-embeddings (Source Cooperative) path, which needs no confirmed key.
- Add the remaining IMPORTANT credentials (NASA Earthdata, TAHMO, WPDx, Twilio) as each module is brought online per the phased roadmap.
- Run `pytest tests/` at any point — the unit tests require no credentials and validate the core scoring/decision logic independently of external services.

Setup reference accompanying the North Rift GeoAI + Agentic EWS production codebase. Keep the populated `.env` file out of version control at all times.